

```
import pandas as pd
```

```
df1 = pd.read_csv("German-Credit_1.csv")
df2 = pd.read_csv("German-Credit_2.csv")
```

```
print("Dataset 1")
print(df1.head())
```

```
print("\nDataset 2")
print(df2.head())
```

```
Dataset 1
  OBS  DURATION  NEW_CAR  USED_CAR  FURNITURE  RADIO_TV  EDUCATION  \
0    1         6.0      0.0      0.0        0.0        1.0        0.0
1    2        48.0      0.0      0.0        0.0        1.0        0.0
2    3        12.0      0.0      0.0        0.0        0.0        1.0
3    4        42.0      0.0      0.0        1.0        0.0        0.0
4    5        24.0      1.0      0.0        0.0        0.0        0.0

  RETRAINING  AMOUNT  INSTALL_RATE  ...  PROP_UNKN_NONE  AGE  OTHER_INSTALL  \
0          0.0  1169.0           4.0  ...           0.0  67.0           0.0
1          0.0  5951.0           2.0  ...           0.0  22.0           0.0
2          0.0  2096.0           2.0  ...           0.0  49.0           0.0
3          0.0  7882.0           2.0  ...           0.0  45.0           0.0
4          0.0  4870.0           3.0  ...           1.0  53.0           0.0

  RENT  OWN_RES  NUM_CREDITS  NUM_DEPENDENTS  TELEPHONE  FOREIGN  RESPONSE
0  0.0      1.0           2.0              1.0         1.0      0.0        1.0
1  0.0      1.0           1.0              1.0         0.0      0.0        0.0
2  0.0      1.0           1.0              2.0         0.0      0.0        1.0
3  0.0      0.0           1.0              2.0         0.0      0.0        1.0
4  0.0      0.0           2.0              2.0         0.0      0.0        0.0

[5 rows x 23 columns]
```

```
Dataset 2
  OBS  CHK_ACCT  HISTORY  SAV_ACCT  EMPLOYMENT  PRESENT_RESIDENT  JOB
0    1         0         4         4          4              4      2
1    2         1         2         0          2              2      2
2    3         3         4         0          3              3      1
3    4         0         2         0          3              4      2
4    5         0         3         0          2              4      2
```

```
print("Shape of Dataset 1:", df1.shape)
print("Shape of Dataset 2:", df2.shape)
```

```
Shape of Dataset 1: (1000, 23)
Shape of Dataset 2: (1000, 7)
```

```
print(df1.columns)
```

```
Index(['OBS', 'DURATION', 'NEW_CAR', 'USED_CAR', 'FURNITURE', 'RADIO_TV',
      'EDUCATION', 'RETRAINING', 'AMOUNT', 'INSTALL_RATE', 'CO_APPLICANT',
      'GUARANTOR', 'REAL_ESTATE', 'PROP_UNKN_NONE', 'AGE', 'OTHER_INSTALL',
      'RENT', 'OWN_RES', 'NUM_CREDITS', 'NUM_DEPENDENTS', 'TELEPHONE',
      'FOREIGN', 'RESPONSE'],
      dtype='object')
```

```
import numpy as np
print(df1.select_dtypes(include=np.number).columns)
```

```
Index(['OBS', 'DURATION', 'NEW_CAR', 'USED_CAR', 'FURNITURE', 'RADIO_TV',
      'EDUCATION', 'RETRAINING', 'AMOUNT', 'INSTALL_RATE', 'CO_APPLICANT',
      'GUARANTOR', 'REAL_ESTATE', 'PROP_UNKN_NONE', 'AGE', 'OTHER_INSTALL',
      'RENT', 'OWN_RES', 'NUM_CREDITS', 'NUM_DEPENDENTS', 'TELEPHONE',
      'FOREIGN', 'RESPONSE'],
      dtype='object')
```

```
df1.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 23 columns):
#   Column              Non-Null Count  Dtype
---  -
0   OBS                 1000 non-null  int64
1   DURATION            998 non-null   float64
2   NEW_CAR              999 non-null   float64
3   USED_CAR             999 non-null   float64
4   FURNITURE            999 non-null   float64
5   RADIO_TV             997 non-null   float64
6   EDUCATION            996 non-null   float64
7   RETRAINING           996 non-null   float64
```

```

8  AMOUNT      990 non-null    float64
9  INSTALL_RATE 993 non-null    float64
10 CO_APPLICANT 992 non-null    float64
11 GUARANTOR   993 non-null    float64
12 REAL_ESTATE 993 non-null    float64
13 PROP_UNKN_NONE 993 non-null    float64
14 AGE         990 non-null    float64
15 OTHER_INSTALL 991 non-null    float64
16 RENT        992 non-null    float64
17 OWN_RES     993 non-null    float64
18 NUM_CREDITS 994 non-null    float64
19 NUM_DEPENDENTS 993 non-null    float64
20 TELEPHONE   995 non-null    float64
21 FOREIGN     994 non-null    float64
22 RESPONSE    997 non-null    float64
dtypes: float64(22), int64(1)
memory usage: 179.8 KB

```

```

print(df1.isnull().sum())
print(df2.isnull().sum())

```

```

OBS                0
DURATION           2
NEW_CAR            1
USED_CAR           1
FURNITURE          1
RADIO_TV           3
EDUCATION          4
RETRAINING         4
AMOUNT            10
INSTALL_RATE       7
CO_APPLICANT       8
GUARANTOR          7
REAL_ESTATE        7
PROP_UNKN_NONE     7
AGE               10
OTHER_INSTALL      9
RENT               8
OWN_RES            7
NUM_CREDITS        6
NUM_DEPENDENTS     7
TELEPHONE          5
FOREIGN            6
RESPONSE           3
dtype: int64
OBS                0
CHK_ACCT           0
HISTORY            0
SAV_ACCT           0
EMPLOYMENT         0
PRESENT_RESIDENT   0
JOB                0
dtype: int64

```

```
print(df1['AMOUNT'].mean())
```

```
3273.168686868687
```

```

print("Maximum Age:", df1['AGE'].max())
print("Minimum Age:", df1['AGE'].min())

```

```

Maximum Age: 75.0
Minimum Age: 19.0

```

```

print("Customers with RESPONSE = 1:")
print((df1['RESPONSE'] == 1).sum())

```

```

Customers with RESPONSE = 1:
698

```

```
print(df1['INSTALL_RATE'].value_counts().sort_index())
```

```

INSTALL_RATE
1.0    135
2.0    230
3.0    155
4.0    473
Name: count, dtype: int64

```

```
df1['AMOUNT_PER_DURATION'] = df1['AMOUNT'] / df1['DURATION']
```

```
print(df1[['AMOUNT', 'DURATION', 'AMOUNT_PER_DURATION']].head())
```

```

   AMOUNT  DURATION  AMOUNT_PER_DURATION
0  1169.0        6.0             194.833333

```

1	5951.0	48.0	123.979167
2	2096.0	12.0	174.666667
3	7882.0	42.0	187.666667
4	4870.0	24.0	202.916667

```
filtered = df1[(df1['AGE'] > 40) & (df1['AMOUNT'] > 5000)]
```

```
print(filtered.head())
```

```
print("\nNumber of matching customers:", len(filtered))
```

	OBS	DURATION	NEW_CAR	USED_CAR	FURNITURE	RADIO_TV	EDUCATION	\
3	4	42.0	0.0	0.0	1.0	0.0	0.0	
18	19	24.0	0.0	1.0	0.0	0.0	0.0	
29	30	60.0	0.0	0.0	0.0	0.0	0.0	
42	43	18.0	0.0	0.0	0.0	0.0	0.0	
44	45	48.0	0.0	1.0	0.0	0.0	0.0	

	RETRAINING	AMOUNT	INSTALL_RATE	...	AGE	OTHER_INSTALL	RENT	\
3	0.0	7882.0	2.0	...	45.0	0.0	0.0	
18	0.0	12579.0	4.0	...	44.0	0.0	0.0	
29	1.0	6836.0	3.0	...	63.0	0.0	0.0	
42	0.0	6204.0	2.0	...	44.0	0.0	0.0	
44	0.0	6143.0	4.0	...	58.0	1.0	0.0	

	OWN_RES	NUM_CREDITS	NUM_DEPENDENTS	TELEPHONE	FOREIGN	RESPONSE	\
3	0.0	1.0	2.0	0.0	0.0	1.0	
18	0.0	1.0	1.0	1.0	0.0	0.0	
29	1.0	2.0	1.0	1.0	0.0	0.0	
42	1.0	1.0	2.0	1.0	0.0	1.0	
44	0.0	2.0	1.0	0.0	0.0	0.0	

	AMOUNT_PER_DURATION
3	187.666667
18	524.125000
29	113.933333
42	344.666667
44	127.979167

```
[5 rows x 24 columns]
```

```
Number of matching customers: 60
```

```
sorted_df = df1.sort_values(by='AMOUNT', ascending=False)
```

```
print(sorted_df[['OBS', 'AMOUNT']].head())
```

	OBS	AMOUNT
915	916	18424.0
95	96	15945.0
818	819	15857.0
887	888	15672.0
637	638	15653.0

```
print(df1.groupby('RESPONSE')['AMOUNT'].mean())
```

```
RESPONSE
0.0    3946.831081
1.0    2986.940837
Name: AMOUNT, dtype: float64
```

```
print(df1.groupby('RESPONSE')['AGE'].mean())
```

```
RESPONSE
0.0    33.787162
1.0    36.207792
Name: AGE, dtype: float64
```

```
merged_df = pd.merge(df1, df2, on='OBS')
```

```
print("Merged Shape:", merged_df.shape)
print(merged_df.head())
```

```
Merged Shape: (1000, 30)
```

	OBS	DURATION	NEW_CAR	USED_CAR	FURNITURE	RADIO_TV	EDUCATION	\
0	1	6.0	0.0	0.0	0.0	1.0	0.0	
1	2	48.0	0.0	0.0	0.0	1.0	0.0	
2	3	12.0	0.0	0.0	0.0	0.0	1.0	
3	4	42.0	0.0	0.0	1.0	0.0	0.0	
4	5	24.0	1.0	0.0	0.0	0.0	0.0	

	RETRAINING	AMOUNT	INSTALL_RATE	...	TELEPHONE	FOREIGN	RESPONSE	\
0	0.0	1169.0	4.0	...	1.0	0.0	1.0	
1	0.0	5951.0	2.0	...	0.0	0.0	0.0	
2	0.0	2096.0	2.0	...	0.0	0.0	1.0	
3	0.0	7882.0	2.0	...	0.0	0.0	1.0	

```
4      0.0 4870.0      3.0 ...      0.0      0.0      0.0
```

```

      AMOUNT_PER_DURATION  CHK_ACCT  HISTORY  SAV_ACCT  EMPLOYMENT  \
0      194.833333      0      4      4      4
1      123.979167      1      2      0      2
2      174.666667      3      4      0      3
3      187.666667      0      2      0      3
4      202.916667      0      3      0      2

```

```

      PRESENT_RESIDENT  JOB
0      4      2
1      2      2
2      3      1
3      4      2
4      4      2

```

```
[5 rows x 30 columns]
```

```
corr_matrix = df1.corr(numeric_only=True)
```

```
print(corr_matrix)
```

```

NUM_DEPENDENTS      0.024881 -0.025855  0.104698  0.055555 -0.087126
TELEPHONE            0.000594  0.165170 -0.041186  0.134376 -0.050794
FOREIGN              -0.018974 -0.139411  0.156039 -0.031466 -0.009663
RESPONSE             -0.037210 -0.211943 -0.095439  0.099893 -0.022867
AMOUNT_PER_DURATION -0.013978 -0.125942  0.105883  0.109085  0.005086

      RADIO_TV  EDUCATION  RETRAINING  AMOUNT  INSTALL_RATE  \
OBS          -0.017823 -0.024465 -0.017114  0.007981  0.012601
DURATION      -0.043652  0.003021  0.163451  0.624527  0.073696
NEW_CAR        -0.345212 -0.127044 -0.181519 -0.038799 -0.048154
USED_CAR       -0.211587 -0.078079 -0.111558  0.255915 -0.089643
FURNITURE      -0.291601 -0.107611 -0.153752 -0.037997 -0.061209
RADIO_TV        1.000000 -0.143411 -0.204903 -0.171473  0.137485
EDUCATION       -0.143411  1.000000 -0.075517 -0.007574  0.042655
RETRAINING     -0.204903 -0.075517  1.000000  0.100059 -0.028608
AMOUNT          -0.171473 -0.007574  0.100059  1.000000 -0.272092
INSTALL_RATE     0.137485  0.042655 -0.028608 -0.272092  1.000000
CO_APPLICANT    -0.005064 -0.047837 -0.033706  0.079098 -0.012870
GUARANTOR        0.115208 -0.054130 -0.046308 -0.065664 -0.006411
REAL_ESTATE      0.127668 -0.103359  0.014621 -0.251932 -0.036792
PROP_UNKN_NONE   -0.118435  0.155794 -0.036604  0.249812  0.045237
AGE              -0.042707  0.060078 -0.000748  0.032818  0.052671
OTHER_INSTALL    -0.033993  0.010185  0.096998  0.048892  0.014848
RENT             -0.074847  0.003123 -0.010055 -0.021982 -0.086714
OWN_RES          0.137808 -0.102747  0.050609 -0.121523  0.047259
NUM_CREDITS      -0.029207 -0.007550  0.085577  0.021438  0.016894
NUM_DEPENDENTS   -0.083941  0.042841 -0.001461  0.016582 -0.070322
TELEPHONE        -0.080115  0.017160  0.088925  0.281637  0.015276
FOREIGN          -0.063830 -0.045300 -0.046820 -0.050463 -0.090008
RESPONSE         0.108934 -0.071132 -0.037365 -0.155549 -0.073559
AMOUNT_PER_DURATION -0.132814 -0.026190 -0.026646  0.522792 -0.395419

      ...      AGE  OTHER_INSTALL  RENT  OWN_RES  \
OBS      ... -0.000972      -0.002021  0.025078 -0.016263
DURATION  ... -0.036725      0.063064 -0.060467 -0.075505
NEW_CAR    ...  0.070218      -0.032474 -0.008678 -0.008677
USED_CAR   ...  0.059165      -0.008875  0.032944 -0.138206
FURNITURE  ... -0.129738      -0.002799  0.109966 -0.045252
RADIO_TV   ... -0.042707      -0.033993 -0.074847  0.137808
EDUCATION  ...  0.060078      0.010185  0.003123 -0.102747
RETRAINING ... -0.000748      0.096998 -0.010055  0.050609
AMOUNT     ...  0.032818      0.048892 -0.021982 -0.121523
INSTALL_RATE ...  0.052671      0.014848 -0.086714  0.047259
CO_APPLICANT ... -0.017675      -0.008605  0.048552 -0.036298
GUARANTOR  ... -0.023270      0.049855  0.020232  0.029046
REAL_ESTATE ...  0.014755      -0.059074  0.029198  0.120161
PROP_UNKN_NONE ...  0.209750      0.060477 -0.068006 -0.476532
AGE        ...  1.000000      0.033570 -0.207242  0.002223
OTHER_INSTALL ...  0.033570      1.000000 -0.067898  0.028735
RENT       ... -0.207242      -0.067898  1.000000 -0.735809
OWN_RES    ...  0.002223      0.028735 -0.735809  1.000000
NUM_CREDITS ...  0.144662      0.048388 -0.054410  0.040786
NUM_DEPENDENTS ...  0.121866      0.071561 -0.062984 -0.028407
TELEPHONE  ...  0.148095      0.021850 -0.051234 -0.033216
FOREIGN    ... -0.005121      -0.026150  0.033233  0.018745
RESPONSE   ...  0.097908      -0.110546 -0.092021  0.131695
AMOUNT_PER_DURATION ...  0.102528      -0.010821  0.018725 -0.056195

```

```
NUM_CREDITS  NUM_DEPENDENTS  TELEPHONE  FOREIGN  \
```

```
import numpy as np
```

```

arr = np.array([
    [1,2,3,4],
    [5,6,7,8],
    [9,10,11,12]
])

```

```
print(arr)
print("Shape:", arr.shape)
print("Size:", arr.size)
print("Dtype:", arr.dtype)
```

```
[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]
Shape: (3, 4)
Size: 12
Dtype: int64
```

```
print(np.zeros((3,3)))
print()

print(np.ones((3,3)))
print()

print(np.full((3,3), 7))
```

```
[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]

[[1. 1. 1.]
 [1. 1. 1.]
 [1. 1. 1.]]

[[7 7 7]
 [7 7 7]
 [7 7 7]]
```

```
print(np.arange(1,11))
print()
print(np.linspace(0,10,5))
```

```
[ 1  2  3  4  5  6  7  8  9 10]

[ 0.   2.5  5.   7.5 10. ]
```

```
np.random.seed(42)

random_arr = np.random.rand(3,3)

print(random_arr)
```

```
[[0.37454012 0.95071431 0.73199394]
 [0.59865848 0.15601864 0.15599452]
 [0.05808361 0.86617615 0.60111501]]
```

```
arr = np.array([10,20,30,40,50])

print("First:", arr[0])
print("Last:", arr[-1])
print("Specific:", arr[2])
```

```
First: 10
Last: 50
Specific: 30
```

```
arr = np.array([
    [1,2,3],
    [4,5,6],
    [7,8,9]
])

submatrix = arr[0:2, 1:3]

print(submatrix)
```

```
[[2 3]
 [5 6]]
```

```
arr = np.arange(1,13)

reshaped = arr.reshape(3,4)

print(reshaped)
```

```
[[ 1  2  3  4]
 [ 5  6  7  8]]
```

```
[ 9 10 11 12]]
```

```
matrix = np.array([
    [1,2,3],
    [4,5,6]
])

print(matrix.T)
```

```
[[1 4]
 [2 5]
 [3 6]]
```

```
arr = np.array([1,2,3,4])

view_arr = arr.view()
copy_arr = arr.copy()

arr[0] = 100

print("Original:", arr)
print("View:", view_arr)
print("Copy:", copy_arr)
```

```
Original: [100  2  3  4]
View: [100  2  3  4]
Copy: [1 2 3 4]
```

```
arr = np.array([1,2,3,4,5])

print(arr[::-1])
```

```
[5 4 3 2 1]
```